

Fist Order Metric Reconstruction in Schwarzschild spacetime

Vahid Toomani^{1,*}

¹*Institut für Theoretische Physik,
Universität Leipzig, Leipzig, Germany and
Max-Planck-Institute MiS, Leipzig, Germany*

(Dated: August 24, 2026)

Abstract

My notes!

* vahid.toomani.2@uni-leipzig.de

CONTENTS

I. GHP & Held	3
II. Components of Homogenous Φ	4
III. Sphroidal Expansions	6
IV. Schwarzschild Case	7
References	9

I. GHP & HELD

Combining eq. 123, eq. 124 and eq. J.1a of [1] gives

$$\zeta = -\Psi_2^{-1/6}\bar{\Psi}_2^{-1/6}\rho^{-1/2}\bar{\rho}^{1/2} \quad (1)$$

$$\begin{aligned} B_a &= -\rho n_a + \tau \bar{m}_a, \\ B'_a &= -\rho' l_a + \tau' m_a. \end{aligned} \quad (2)$$

$$t_a = \zeta(B_a - B'_a) \quad (3)$$

$$\mathbb{L}_t \eta = t^a \Theta_a \eta - \frac{p}{2} \zeta \Psi_2 \eta - \frac{q}{2} \bar{\zeta} \bar{\Psi}_2 \eta = \zeta \rho' \mathbb{P} \eta - \zeta \rho \mathbb{P}' \eta - \zeta \tau' \bar{\partial} \eta + \zeta \tau \bar{\partial}'' \eta - \frac{p}{2} \zeta \Psi_2 \eta - \frac{q}{2} \bar{\zeta} \bar{\Psi}_2 \eta \quad (4)$$

$$\begin{aligned} \mathbb{P} \mathbb{P} \mathbb{P} \mathbb{P} (\zeta^4 \psi_4) &= \bar{\partial}' \bar{\partial}' \bar{\partial}' \bar{\partial}' (\zeta^4 \psi_0) - 3 \mathbb{L}_t \bar{\psi}_0, \\ \mathbb{P}' \mathbb{P}' \mathbb{P}' \mathbb{P}' (\zeta^4 \psi_0) &= \bar{\partial} \bar{\partial} \bar{\partial} \bar{\partial} (\zeta^4 \psi_4) + 3 \mathbb{L}_t \bar{\psi}_4. \end{aligned} \quad (5)$$

II. COMPONENTS OF HOMOGENOUS Φ

In this section, I express how one achieves a metric perturbation that falls fast enough at $\mathcal{I}^+$. To achieve this end I used the procedure demonstrated in details by Hollands et. al. in [1]. This method is based on the GHZ reconstruction [2] combined with the puncture scheme [3]. The puncture scheme implemented in [3] breaks the retarded solution h_{ab}^{ret} into two parts of punctured field h_{ab}^{P} which contains the singular field and the residual part h_{ab}^{Res} . The residual field satisfy a homogeneous linearized Einstein equation sourced by $\mathcal{E}h_{ab}^{\text{P}}$ away from the singular source, i.e.,

$$\mathcal{E}h_{ab}^{\text{Res}} = -\mathcal{E}h_{ab}^{\text{P}}. \quad (6)$$

The puncture field is an estimate for the singular field in a neighborhood of the orbiting compact object. This estimate is given by the product of the singular field and a $W(r, \theta, \phi)$ with the following properties:

1. W is compactly supported (usually $W = 0$ for $r < r_{\min}$ and $r > r_{\max}$, the support of W is called puncture region;
2. $W = 1$ on the location of the compact object.

One obvious choice would be $W(r) = \Theta(r - r_{\min}) - \Theta(r_{\max} - r)$ but one might choose a smooth one too. Consequently, residual field h_{ab}^{Res} is only sourced in the puncture region.

Applying $\mathcal{S}$ on both sides of (6) and using the Wald identity [?],

$$\mathcal{O}\psi_0^{\text{Res}} = -\mathcal{S}^{ab}T_{ab}^{\text{P}} = -T_0^{\text{P}}. \quad (7)$$

Using the Wald identity [?], we have

Combining eq. 123, eq. 124 and eq. J.1a of [1] gives

$$2\tilde{\mathbf{P}}'\tilde{\mathbf{P}}'\tilde{\mathbf{P}}'\bar{\psi}_0^{5\circ} = \tilde{\delta}'\tilde{\delta}'\tilde{\delta}'\tilde{\delta}'\tilde{\delta}\tilde{\delta}\tilde{\delta}\tilde{\delta}\bar{f}^\circ - 9\Psi^\circ\bar{\Psi}^\circ\tilde{\mathbf{P}}'\tilde{\mathbf{P}}'\bar{f}^\circ. \quad (8)$$

By expanding $\bar{f}^\circ$ and $\bar{\psi}_0^{5\circ}$ in spheroidal (spin-weighted harmonics in Schwarzschild) harmonics, one can reverse this equation and obtain $\bar{f}^\circ$. Next, we obtain following Held-like scalars $\Phi^{i\circ}$ from $\bar{f}^\circ$ (and potentially ODEs in terms of $\frac{\partial}{\partial u}$).

$$\tilde{\mathbf{P}}'\tilde{\mathbf{P}}'\Phi^{0\circ} = -\frac{1}{3}\tilde{\delta}'\tilde{\delta}'\tilde{\delta}'\tilde{\delta}\tilde{\delta}\tilde{\delta}\bar{f}^\circ - \left[2\bar{\tau}^\circ\tilde{\delta}'\tilde{\delta}\tilde{\delta} + \Psi^\circ\tilde{\delta}'\tilde{\delta} + 2\Psi^\circ\rho'^\circ\right]\tilde{\mathbf{P}}'\bar{f}^\circ \quad (9)$$

$$\tilde{\mathbf{P}}'\Phi^{1\circ} = -\tilde{\delta}'\tilde{\delta}'\tilde{\delta}\tilde{\delta}\bar{f}^\circ - \left[4\bar{\tau}^\circ\tilde{\delta} + 3\Psi^\circ\right]\tilde{\mathbf{P}}'\bar{f}^\circ \quad (10)$$

$$\Phi^{2\circ} = -2\tilde{\delta}'\tilde{\delta}\bar{f}^\circ, \quad (11)$$

$$\Phi^{3\circ} = -2\tilde{\mathbf{P}}'\bar{f}^\circ, \quad (12)$$

$$\bar{\zeta}_m^\circ = \tilde{\delta}\bar{f}^\circ, \quad (13)$$

$$\tilde{\mathbf{P}}'\zeta_l^\circ = -\text{Re}\left\{\tilde{\delta}\tilde{\delta}\bar{f}^\circ\right\}, \quad (14)$$

$$\tilde{\mathbf{P}}'\zeta_n^\circ = -\text{Re}\left\{\tilde{\delta}'\tilde{\delta}\tilde{\delta}\bar{f}^\circ - \rho'^\circ\tilde{\delta}\tilde{\delta}\bar{f}^\circ - 2\tau^\circ\tilde{\delta}\tilde{\mathbf{P}}'\bar{f}^\circ - \frac{1}{2}\Omega^\circ\tilde{\delta}\tilde{\delta}\tilde{\mathbf{P}}'\bar{f}^\circ\right\}, \quad (15)$$

kslfhadlf

$$\begin{aligned} h_{ab} &\rightarrow h_{ab} - \mathcal{L}_\zeta g_{ab} \\ \Phi &\rightarrow \Phi + \Phi^{0\circ} + \Phi^{1\circ}\frac{1}{\rho} + \Phi^{2\circ}\frac{1}{\rho^2} + \Phi^{3\circ}\frac{1}{\rho^3} \end{aligned} \quad (16)$$

$$\begin{aligned} \mathcal{L}_\zeta g_{ab}l^a &= 0 \\ \mathcal{L}_\zeta g_{ab}m^a\bar{m}^b &= 0 \end{aligned} \quad (17)$$

$$\begin{aligned}
\mathcal{L}_\zeta g_{ab} n^a n^b = \text{Re} \Bigg[& 4\rho\bar{\rho}\bar{\tau}^\circ\tilde{\partial}'\tilde{\partial}\tilde{\partial}\zeta_l^\circ - 4\rho^2\bar{\rho}\bar{\tau}^{\circ 2}\tilde{\partial}\tilde{\partial}\zeta_l^\circ + 2\rho^2\Psi^\circ\tilde{\partial}'\tilde{\partial}\zeta_l^\circ + 4\rho^2\bar{\rho}\Psi^\circ\bar{\tau}^\circ\tilde{\partial}\zeta_l^\circ \\
& - 2\tilde{\partial}'\tilde{\partial}'\tilde{\partial}'\tilde{\partial}f^\circ - 2\left(\frac{1}{\rho} + 4\Omega^\circ\right)\tilde{\partial}'\tilde{\partial}'\tilde{\mathbf{P}}'f^\circ - 4\rho\bar{\rho}\Omega^\circ\bar{\tau}^\circ\tilde{\partial}'\tilde{\partial}'\tilde{\partial}f^\circ + 4\left(4 + \bar{\rho}\Omega^\circ - \rho\bar{\rho}\Omega^{\circ 2}\right)\bar{\tau}^\circ\tilde{\partial}'\tilde{\mathbf{P}}'f^\circ \\
& - \left(8\rho'^\circ + 3\rho\Psi^\circ + 3\bar{\rho}\bar{\Psi}^\circ + 2\rho^2\Omega^{\circ 2}\Psi^\circ + 8\rho\bar{\rho}\tau^\circ\bar{\tau}^\circ\right)\tilde{\partial}'\tilde{\partial}'f^\circ + 4\rho^2\bar{\tau}^{\circ 2}\tilde{\partial}'\tilde{\partial}f^\circ \\
& + 4\rho(\rho + 2\bar{\rho})\Omega^\circ\bar{\tau}^{\circ 2}\tilde{\mathbf{P}}'f^\circ + 8\rho^2\bar{\rho}\tau^\circ\bar{\tau}^{\circ 2}\tilde{\partial}'f^\circ + 4\rho\bar{\rho}\bar{\Psi}^\circ\bar{\tau}^\circ\tilde{\partial}'f^\circ - 24\rho\bar{\rho}\Omega^\circ\rho'^\circ\bar{\tau}^\circ\tilde{\partial}'f^\circ + 16\rho^2\rho'^\circ\bar{\tau}^{\circ 2}f^\circ \Bigg]
\end{aligned} \tag{18}$$

$$\begin{aligned}
\mathcal{L}_\zeta g_{ab} n^a m^b = & \bar{\rho}\tilde{\partial}'\tilde{\partial}\tilde{\partial}\zeta_l^\circ - 2\rho\bar{\rho}\bar{\tau}\tilde{\partial}\tilde{\partial}\zeta_l^\circ + \rho(\rho + \bar{\rho})\Psi^\circ\tilde{\partial}\zeta_l^\circ - \bar{\rho}\Omega^\circ\tilde{\partial}'\tilde{\partial}'\tilde{\partial}f^\circ + 2\rho\bar{\tau}^\circ\tilde{\partial}'\tilde{\partial}f^\circ \\
& + \left(\frac{1}{\bar{\rho}} - 2\bar{\rho}\Omega^{\circ 2}\right)\tilde{\partial}'\tilde{\mathbf{P}}'f^\circ + 2(\rho + \bar{\rho})\Omega^\circ\bar{\tau}^\circ\tilde{\mathbf{P}}'f^\circ - 2\bar{\rho}\tau^\circ\tilde{\partial}'\tilde{\partial}'f^\circ - (\rho^2\Omega^\circ + \rho)\Psi^\circ\tilde{\partial}'f^\circ \\
& + \bar{\rho}\bar{\Psi}^\circ\tilde{\partial}'f^\circ - 6\bar{\rho}\Omega^\circ\rho'^\circ\tilde{\partial}'f^\circ + 4\rho\bar{\rho}\tau^\circ\bar{\tau}^\circ\tilde{\partial}'f^\circ + 8\rho\rho'^\circ\bar{\tau}^\circ f^\circ
\end{aligned} \tag{19}$$

$$\mathcal{L}_\zeta g_{ab} m^a m^b = -2\bar{\rho}\tilde{\partial}\tilde{\partial}\zeta_l^\circ + 2\tilde{\partial}'\tilde{\partial}f^\circ + 2\Omega^\circ\tilde{\mathbf{P}}'f^\circ + 4\bar{\rho}\tau^\circ\tilde{\partial}'f^\circ + 8\rho'^\circ f^\circ \tag{20}$$

III. SPHROIDAL EXPANSIONS

$$\psi_4^{1\circ} = \sum_{\ell m} \int d\omega \psi_{4,\ell m}^{1\circ}(\omega) {}_{-2}S_{\ell m}(\theta, \varphi_\star; a\omega) e^{-i\omega u} \tag{21}$$

$$\psi_0^{5\circ} = \sum_{\ell m} \int d\omega \psi_{0,\ell m}^{5\circ}(\omega) {}_{+2}S_{\ell m}(\theta, \varphi_\star; a\omega) e^{-i\omega u} \tag{22}$$

$$\begin{aligned}
\psi_{4,\ell m}^{1\circ}(\omega) = & \frac{\omega^4}{-{}_2B_{\ell m}(a\omega) {}_{-2}\bar{B}_{\ell-m}(-a\omega) + 9M^2\omega^2} \\
& \left[{}_{-2}\bar{B}_{\ell-m}(-a\omega)\psi_{0,\ell m}^{5\circ}(\omega) + 3i\omega(-1)^m\bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) \right]
\end{aligned} \tag{23}$$

$$\bar{h}_{mm}^{1\circ} = \sum_{\ell m} \int d\omega \frac{-2\omega^2}{-{}_2B_{\ell m}(a\omega) - {}_2\bar{B}_{\ell-m}(-a\omega) + 9M^2\omega^2} \times \\ \left[-{}_2\bar{B}_{\ell-m}(-a\omega)\psi_{0,\ell m}^{5\circ}(\omega) + 3i\omega(-1)^m\bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) \right] {}_2S_{\ell m}(\theta, \varphi_\star; a\omega)e^{-i\omega u} \quad (24)$$

$$f^\circ = \sum_{\ell m} \int d\omega \frac{2i\omega^3}{-{}_2B_{\ell m}(a\omega) - {}_2\bar{B}_{\ell-m}(-a\omega) + 9M^2\omega^2} \psi_{0,\ell m}^{5\circ}(\omega) {}_2S_{\ell m}(\theta, \varphi_\star; a\omega)e^{-i\omega u} \quad (25)$$

$$e^\circ = \sum_{\ell m} \int d\omega \frac{-4\omega^4(-1)^m}{-{}_2B_{\ell m}(a\omega) - {}_2\bar{B}_{\ell-m}(-a\omega) + 9M^2\omega^2} \bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) {}_2S_{\ell m}(\theta, \varphi_\star; a\omega)e^{-i\omega u} \quad (26)$$

IV. SCHWARZSCHILD CASE

$$-{}_2B_{\ell-m}(a\omega) = {}_2B_\ell = \frac{1}{4} \frac{(\ell+2)!}{(\ell-2)!} = \frac{1}{4}(\ell+2)(\ell+1)\ell(\ell-1) \quad (27)$$

$$-{}_1B_{\ell-m}(a\omega) = {}_1B_\ell = \frac{1}{2} \frac{(\ell+1)!}{(\ell-1)!} = \frac{1}{2}\ell(\ell+1) \quad (28)$$

$$\Phi^{0\circ} = d^\circ = \sum_{\ell m} \int d\omega \frac{-2/3i\omega(-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} \left[{}_2B_\ell {}_1B_\ell - 3iM\omega(1 + \sqrt{{}_2B_\ell}) \right] \bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) {}_2Y_{\ell m}(\theta, \varphi_\star)e^{-i\omega u} \quad (29)$$

$$\Phi^{1\circ} = \sum_{\ell m} \int d\omega \frac{2\omega^2(-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} [{}_2B_\ell - 3iM\omega] \bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) {}_2Y_{\ell m}(\theta, \varphi_\star)e^{-i\omega u} \quad (30)$$

$$\Phi^{2\circ} = \sum_{\ell m} \int d\omega \frac{4i\omega^3 {}_2B_\ell / {}_1B_\ell (-1)^m}{{}_2B_\ell^2 + 9M^2\omega^2} \bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) {}_2Y_{\ell m}(\theta, \varphi_\star)e^{-i\omega u} \quad (31)$$

$$\Phi^{3\circ} = e^\circ = \sum_{\ell m} \int d\omega \frac{-4\omega^4(-1)^m}{2B_\ell^2 + 9M^2\omega^2} \bar{\psi}_{0,\ell-m}^{5\circ}(-\omega) {}_{-2}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \quad (32)$$

$$\zeta_m^\circ = \sum_{\ell m} \int d\omega \frac{-2i\omega^3 \sqrt{{}_2B_\ell / {}_1B_\ell}}{2B_\ell^2 + 9M^2\omega^2} \psi_{0,\ell m}^{5\circ}(\omega) {}_{+1}Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} \quad (33)$$

$$\zeta_l^\circ = \sum_{\ell m} \int d\omega \frac{\omega^2 \sqrt{{}_2B_\ell}}{2B_\ell^2 + 9M^2\omega^2} \psi_{0,\ell m}^{5\circ}(\omega) {}_0Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} + cc. \quad (34)$$

$$\zeta_n^\circ = \sum_{\ell m} \int d\omega \frac{-\omega^2 \sqrt{{}_2B_\ell}}{2B_\ell^2 + 9M^2\omega^2} \left({}_1B_\ell + \frac{1}{2} \right) \psi_{0,\ell m}^{5\circ}(\omega) {}_0Y_{\ell m}(\theta, \varphi_\star) e^{-i\omega u} + cc. \quad (35)$$

-
- [1] S. Hollands and V. Toomani, Metric reconstruction in Kerr spacetime, *Class. Quant. Grav.* **43**, 055001 (2026), arXiv:2405.18604 [gr-qc].
 - [2] S. R. Green, S. Hollands, and P. Zimmerman, Teukolsky formalism for nonlinear Kerr perturbations, *Class. Quant. Grav.* **37**, 075001 (2020), arXiv:1908.09095 [gr-qc].
 - [3] V. Toomani, P. Zimmerman, A. Spiers, S. Hollands, A. Pound, and S. R. Green, New metric reconstruction scheme for gravitational self-force calculations, *Class. Quant. Grav.* **39**, 015019 (2022), arXiv:2108.04273 [gr-qc].